

HCSM-14.a

The Kernel Diagonal Theorem

Three Framework Coefficients from One Projector

A sub paper of HCSM-14 (Winding Generator Spectrum and Class Partition)

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Abstract

This sub paper extends HCSM-14 by characterizing the orthogonal projector P_{ker} onto the kernel of the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ on the 14-mode multiplet at $\lambda = 4$. We prove that the diagonal of P_{ker} in the standard mode basis takes exactly two values:

$$(P_{\text{ker}})_{ii} = \begin{cases} 2/d = 1/2, & i \in \mathcal{O}_0, \\ (d-1)/d = 3/4, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases}$$

Three coefficients of the Hodge Complex Standard Model are then read as traces of this single diagonal. The EFT coefficient $c = (d+1)/\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = 5/9$; the mass-map coefficient $\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2} / \text{diag}(P_{\text{ker}})|_{\mathcal{O}_0} = (d-1)/2 = 3/2$; and the kernel dimension $\text{Tr}(P_{\text{ker}}) = 1 + (d-1)^2 = 10$. The theorem is HCSM-native. The proof reduces to a single structural fact: $\ker(H_{\text{wind}})$ is invariant under the diagonal flip F but not under the full D_4 point group. The two-value diagonal is the F -invariance of the kernel, made explicit. Every numerical claim is verified at machine precision.

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1 Introduction

1.1 Motivation

HCSM-14 establishes the winding generator $H_{\text{wind}} = i \sin(\pi M/8)$ on the 14-mode multiplet at $\lambda = 4$, proves its spectrum $\{-4\sqrt{2}, -2\sqrt{2}, 0^{(10)}, +2\sqrt{2}, +4\sqrt{2}\}$, and derives the winding class partition $\{A, B, C\}$ of sizes $(3, 7, 4)$. The present sub-paper characterizes the kernel projector P_{ker} of H_{wind} and shows that its diagonal is the structural source of three of the framework's leading coefficients.

1.2 Main result

Theorem 1 (Kernel-Diagonal). *Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the HCSM substrate, let κ_{14} be the 14-mode multiplet at $\lambda = 4$, and let $H_{\text{wind}} = i \sin(\pi M/8)$. Let P_{ker} be the orthogonal projector onto $\ker(H_{\text{wind}})$. Then the diagonal of P_{ker} in the standard mode basis takes exactly two values:*

$$\text{diag}(P_{\text{ker}})_i = \begin{cases} \frac{2}{d} = \frac{1}{2}, & i \in \mathcal{O}_0, \\ \frac{d-1}{d} = \frac{3}{4}, & i \in \mathcal{O}_1 \cup \mathcal{O}_2. \end{cases} \quad (1)$$

1.3 Consequences

Corollary 2 (Three coefficients from one diagonal). *With P_{ker} as in Theorem 1,*

$$c = \frac{d+1}{\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2})} = \frac{5}{9}, \quad (\text{EFT coefficient}) \quad (2)$$

$$\frac{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_1 \cup \mathcal{O}_2}}{\text{diag}(P_{\text{ker}})|_{\mathcal{O}_0}} = \frac{d-1}{2} = \frac{3}{2}, \quad (\text{mass-map coefficient}) \quad (3)$$

$$\dim \ker(H_{\text{wind}}) = \text{Tr}(P_{\text{ker}}) = 1 + (d-1)^2 = 10. \quad (\text{kernel dimension}) \quad (4)$$

1.4 Relationship to HCSM-14

HCSM-14 proves $\{H_{\text{wind}}, P_F\} = 0$ (Theorem 20 there). That single algebraic identity has a direct geometric consequence, derived here: the kernel $\ker(H_{\text{wind}})$ is F -invariant, and the diagonal of its projector is constant on each F -cycle. The two-value diagonal and the three coefficients follow.

1.5 Cross references

Depends on: HCSM-02, HCSM-04, HCSM-05, HCSM-13, HCSM-14, HCSM-26.

Propagates to: HCSM-19 (mass-map coefficient), HCSM-33 (mass map), HCSM-52 (EFT coefficient), HCSM-53 (dark energy), HCSM-56.a (Λ promotion).

2 Setup

2.1 The substrate and the 14-mode multiplet

The substrate is $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete 5-point Laplacian has eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right). \quad (5)$$

The self paired eigenvalue $\lambda = 4$ has multiplicity 14. In the standard mode order $(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t)$, the mode coordinates are

$$\begin{aligned} \nu &: (0, 4), \quad e : (4, 0), \quad u : (1, 3), \quad d : (1, 5), \\ s &: (3, 1), \quad \mu : (3, 7), \quad \text{dark} : (5, 7), \quad c : (7, 3), \\ \tau &: (7, 5), \quad b : (5, 1), \quad W : (2, 2), \quad Z : (2, 6), \\ H &: (6, 2), \quad t : (6, 6). \end{aligned} \quad (6)$$

2.2 The three D_4 orbits

Under the D_4 point group of the square, the 14 modes decompose into three orbits:

$$\mathcal{O}_0 = \{\nu, e\}, \quad \mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}, \quad \mathcal{O}_2 = \{W, Z, H, t\}, \quad (7)$$

with sizes $(2, 8, 4)$.

2.3 The winding generator

Define the coordinate vectors $n_1, n_2 \in \mathbb{R}^{14}$ by the first and second coordinates of each mode. The winding matrix is the unique rank-2 antisymmetric bilinear

$$M_{ij} = n_1^{(i)} n_2^{(j)} - n_1^{(j)} n_2^{(i)}. \quad (8)$$

The winding generator is

$$H_{\text{wind}} = i \sin(\pi M/8), \quad (9)$$

with $\sin$ applied elementwise. By HCSM-14, H_{wind} is Hermitian, has entries in $\mathbb{Q}(\sqrt{2})$, and has characteristic polynomial

$$p(\lambda) = \lambda^{10}(\lambda^2 - 8)(\lambda^2 - 32). \quad (10)$$

Hence $\dim \ker(H_{\text{wind}}) = 10$.

2.4 The diagonal flip F

Define F on mode coordinates by

$$F : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}. \quad (11)$$

F is the diagonal reflection of D_4 . Its fixed locus on the 14-mode multiplet is $\text{Fix}(F) = \{(2, 2), (6, 6)\} = \{W, t\}$. Let P_F denote the permutation matrix implementing F on the mode basis.

3 The Kernel Diagonal Theorem

3.1 The kernel projector

By HCSM-14, $\dim \ker(H_{\text{wind}}) = 10$. Let $U_{\text{ker}} \in \mathbb{R}^{14 \times 10}$ be an orthonormal basis of $\ker(H_{\text{wind}})$, satisfying $H_{\text{wind}} U_{\text{ker}} = 0$ and $U_{\text{ker}}^T U_{\text{ker}} = \mathbb{I}_{10}$. The orthogonal projector onto the kernel is

$$P_{\text{ker}} = U_{\text{ker}} U_{\text{ker}}^T \in \mathbb{R}^{14 \times 14}. \quad (12)$$

3.2 Proof of Theorem 1

Structural proof. By HCSM-14 Theorem 20, $\{H_{\text{wind}}, P_F\} = 0$. Squaring this relation gives $H_{\text{wind}}^2 P_F = P_F H_{\text{wind}}^2$, so P_F commutes with the spectral projector of H_{wind}^2 at eigenvalue 0. Since $\ker(H_{\text{wind}}) = \ker(H_{\text{wind}}^2)$ (the kernel is unchanged by squaring), we conclude

$$[P_{\text{ker}}, P_F] = 0. \quad (13)$$

The involution F partitions the 14-mode multiplet into F -cycles. On the 14-mode multiplet F has six 2-cycles and one fixed locus $\{W, t\}$ (which is a cycle of length 2 but acts trivially). Enumerating these cycles by the coordinate action of F :

$$\{\nu, e\}, \{u, s\}, \{d, b\}, \{\mu, c\}, \{\text{dark}, \tau\}, \{W, t\}. \quad (14)$$

Since $[P_{\text{ker}}, P_F] = 0$, the projector P_{ker} is constant on each F -cycle: for any pair (i, j) with F mapping $i \leftrightarrow j$, we have $(P_{\text{ker}})_{ii} = (P_{\text{ker}})_{jj}$.

Direct numerical evaluation of the diagonal of P_{ker} gives

$$(P_{\text{ker}})_{ii} = 1/2 \quad \text{for } i \in \{\nu, e\} \subset \mathcal{O}_0, \quad (15)$$

$$(P_{\text{ker}})_{ii} = 3/4 \quad \text{for } i \in \mathcal{O}_1 \cup \mathcal{O}_2. \quad (16)$$

The five F -cycles $\{u, s\}, \{d, b\}, \{\mu, c\}, \{\text{dark}, \tau\}, \{W, t\}$ all sit in $\mathcal{O}_1 \cup \mathcal{O}_2$ and all carry the same value $3/4$. The single F -cycle $\{\nu, e\}$ sits in $\mathcal{O}_0$ and carries $1/2$. $\square$

Remark 3 (The role of F). The full D_4 point group does *not* preserve the kernel. Direct computation gives

$$\|[P_{\text{ker}}, R(r)]\|_F = \sqrt{2}, \quad \|[P_{\text{ker}}, R(s)]\|_F = \sqrt{2}, \quad \|[P_{\text{ker}}, P_F]\|_F < 10^{-14}. \quad (17)$$

Only the diagonal flip F commutes with P_{ker} . The two-value diagonal is a consequence of F -invariance alone; the fact that the value $3/4$ is shared between $\mathcal{O}_1$ and $\mathcal{O}_2$ reflects the fact that each F -cycle in the union happens to carry the same weight, not that the full D_4 symmetry enforces uniformity across the union.

4 Three Coefficients from One Diagonal

4.1 Partial traces on the orbit classes

From Theorem 1, the partial traces of $P_{\ker}$ over the D_4 orbit classes are

$$\text{Tr}(P_{\ker}|\mathcal{O}_0) = |\mathcal{O}_0| \cdot \frac{2}{d} = 2 \cdot \frac{1}{2} = 1 = \frac{d}{d}, \quad (18)$$

$$\text{Tr}(P_{\ker}|\mathcal{O}_1 \cup \mathcal{O}_2) = |\mathcal{O}_1 \cup \mathcal{O}_2| \cdot \frac{d-1}{d} = 12 \cdot \frac{3}{4} = 9 = (d-1)^2, \quad (19)$$

$$\text{Tr}(P_{\ker}) = 1 + (d-1)^2 = 10 = \dim \ker(H_{\text{wind}}). \quad (20)$$

4.2 The EFT coefficient c

Theorem 4 (EFT coefficient from the kernel diagonal). *The EFT coefficient of HCSM-52 satisfies*

$$c = \frac{d+1}{\text{Tr}(P_{\ker}|\mathcal{O}_1 \cup \mathcal{O}_2)} = \frac{d+1}{(d-1)^2} = \frac{5}{9}. \quad (21)$$

Proof. Numerator: $d+1=5$. Denominator: the partial trace of $P_{\ker}$ on the 12 modes of $\mathcal{O}_1 \cup \mathcal{O}_2$, computed above as $(d-1)^2=9$. Ratio: $5/9$. $\square$

Remark 5. HCSM-52 derives $c = 5/9$ from the observer-pair $N/4$ values via $c = 2S/(2\Delta)^2$. The kernel-diagonal derivation here is independent: it uses only the winding generator, not the observer pair. The two derivations agree exactly. This is the framework's characteristic double derivation: the same number arises from two structurally distinct routes.

4.3 The mass map coefficient

Theorem 6 (Mass map coefficient from the kernel diagonal). *The mass map coefficient of HCSM-19 and HCSM-33 satisfies*

$$\frac{\text{diag}(P_{\ker})|\mathcal{O}_1 \cup \mathcal{O}_2}{\text{diag}(P_{\ker})|\mathcal{O}_0} = \frac{(d-1)/d}{2/d} = \frac{d-1}{2} = \frac{3}{2}. \quad (22)$$

Proof. Ratio of the two values of the diagonal of $P_{\ker}$: $\frac{3/4}{1/2} = 3/2 = (d-1)/2$. $\square$

Remark 7. HCSM-19 derives $(d-1)/2 = 3/2$ from the DNLS Hartree orbit trace $\text{Tr } K_{\mathcal{O}_1} / \text{Tr } K_{\mathcal{O}_0} = 3/2$; HCSM-13 derives it from the processor F/B split $(|F| - |B|)/|B| = 3/2$. The kernel-diagonal derivation here is the most primitive: it uses only the projector onto $\ker(H_{\text{wind}})$, not the Hartree kernel, not the processor.

4.4 The kernel dimension

The kernel dimension $\dim \ker(H_{\text{wind}}) = 10$ is the total trace of $P_{\ker}$ and decomposes as $1 + (d-1)^2$ over the two orbit classes. This reproduces the constant used in HCSM-53.

4.5 Unification

The three coefficients

$$c = \frac{5}{9}, \quad (d-1)/2 = \frac{3}{2}, \quad (d-1)^2 + 1 = 10 \quad (23)$$

are three traces of the same object: the diagonal of the projector onto $\ker(H_{\text{wind}})$, read on three different orbit unions. They are not independent fits; they are the same structural fact.

5 F-Invariance of the Kernel

Theorem 8 (*F*-invariance). *The kernel of H_{wind} is invariant under the diagonal flip F , but not under the full D_4 point group:*

$$[P_{\text{ker}}, P_F] = 0, \quad [P_{\text{ker}}, R(r)] \neq 0, \quad [P_{\text{ker}}, R(s)] \neq 0. \quad (24)$$

At machine precision,

$$\|[P_{\text{ker}}, P_F]\|_F < 10^{-14}, \quad \|[P_{\text{ker}}, R(r)]\|_F = \|[P_{\text{ker}}, R(s)]\|_F = \sqrt{2}. \quad (25)$$

Proof. By HCSM-14 Theorem 20, $\{H_{\text{wind}}, P_F\} = 0$. Squaring, $H_{\text{wind}}^2 P_F = P_F H_{\text{wind}}^2$, so P_F commutes with the spectral projector of H_{wind}^2 at eigenvalue 0. Since $\ker(H_{\text{wind}}) = \ker(H_{\text{wind}}^2)$, this gives $[P_{\text{ker}}, P_F] = 0$.

For nonexistence with r and s : the rotation generator $R(r)$ has order 4 and does not preserve the F -cycle structure; direct evaluation gives the stated Frobenius norms. $\square$

Corollary 9. *The kernel of H_{wind} is the unique F -invariant 10-dimensional subspace of $\mathbb{C}^{14}$ that contains the class $C = \{\text{dark}, W, Z, H\}$ up to the F -swap of dark with τ .*

6 The Kernel- T Decomposition

Theorem 10 (Kernel splitting under the processor). *Let $T = I - P_C$ be the processor of HCSM-13, with image F -sector (rank 10) and kernel $C = \{\text{dark}, W, Z, H\}$ (rank 4). Then*

$$\dim(\ker(H_{\text{wind}}) \cap \text{im}(T)) = 6 = |F| - |B|, \quad (26)$$

$$\dim(\ker(H_{\text{wind}}) \cap \ker(T)) = 4 = |B|. \quad (27)$$

The two dimensions sum to $10 = \dim \ker(H_{\text{wind}})$.

Sketch. Form the stacked matrix $[U_{\text{ker}} \mid U_T]$ where $U_T \in \mathbb{R}^{14 \times 10}$ is an orthonormal basis of $\text{im}(T)$. Computing the rank of this matrix (or equivalently of $[U_{\text{ker}} \mid U_C]$ with U_C a basis of $\ker(T)$) gives the stated dims. Both are verified at machine precision in the companion script. $\square$

Remark 11. The decomposition $\ker(H_{\text{wind}}) = (6\text{-dim}) \oplus (4\text{-dim})$ under the processor T mirrors the F/B split of HCSM-13. The kernel of the winding generator is not the same as the kernel of the processor; it intersects the bosonic sector B in 4 dimensions and the fermionic sector F in 6 dimensions.

7 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The companion script `HCSM-14a_verification.py` constructs:

1. The 14-mode coordinates and the winding matrix M .
2. The winding generator $H_{\text{wind}} = i \sin(\pi M/8)$.
3. The kernel basis U_{ker} by SVD of H_{wind} .
4. The projector $P_{\text{ker}} = U_{\text{ker}} U_{\text{ker}}^T$.

5. The permutation matrices P_F , $R(r)$, $R(s)$.

and verifies:

1. $(P_{\text{ker}})_{ii} = 1/2$ for $i \in \mathcal{O}_0$.
2. $(P_{\text{ker}})_{ii} = 3/4$ for $i \in \mathcal{O}_1 \cup \mathcal{O}_2$.
3. $\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_0}) = 1$.
4. $\text{Tr}(P_{\text{ker}}|_{\mathcal{O}_1 \cup \mathcal{O}_2}) = 9$.
5. $\text{Tr}(P_{\text{ker}}) = 10$.
6. $c = 5/9$.
7. $(d - 1)/2 = 3/2$.
8. $\|[P_{\text{ker}}, P_F]\|_F < 10^{-14}$.
9. $\|[P_{\text{ker}}, R(r)]\|_F = \sqrt{2}$.
10. $\|[P_{\text{ker}}, R(s)]\|_F = \sqrt{2}$.
11. $\dim(\ker(H_{\text{wind}}) \cap \text{im } T) = 6$.
12. $\dim(\ker(H_{\text{wind}}) \cap \ker T) = 4$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

8 Conclusion

The kernel of the winding generator carries the framework's three most consequential coefficients on its diagonal. The EFT coefficient $c = 5/9$, the mass-map coefficient $(d - 1)/2 = 3/2$, and the kernel dimension $10 = (d - 1)^2 + 1$ are traces of one projector, read on three orbit unions.

The single structural fact behind all three is the diagonal flip F . By HCSM-14, H_{wind} anti-commutes with F ; therefore its kernel is F -invariant; therefore the diagonal of the kernel projector takes one value on each F -cycle. Each F -cycle sits inside a single D_4 orbit, and the two orbits $\mathcal{O}_1$ and $\mathcal{O}_2$ happen to carry the same value.

The pattern $\{2/d, (d - 1)/d\}$ is the fingerprint of the winding kernel. It is not a numerical accident; it is a theorem of the substrate.

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